But 30 Hiroshimas was not to be sneezed at, nor was deliverability. Joe-4 was an awesome weapon indeed, and Wheeler and other Americans heaved a sigh of relief that, thanks to their own, true superbomb, the new Soviet leader, Georgi Malenkov, could not threaten America with it.

On 1 March 1954, the Americans exploded their first LiD-fueled, deliverable superbomb. It was code named *Bravo* and like Mike, it relied on design calculations by the Wheeler and Los Alamos teams and used the Teller–Ulam invention. The explosive energy was 1300 Hiroshimas.

In March 1954, Sakharov and Zel'dovich jointly invented (independently of the Americans) the Teller–Ulam idea, and within a few months Soviet resources were focused on implementing it in a real superbomb, one that could have as large a destructive power as anyone might wish. It took just eighteen months to fully design and construct the bomb. On 23 November 1955, it was detonated, with an explosive energy of 300 Hiroshimas.

As Oppenheimer's General Advisory Committee had suspected, in their opposition to the crash program for the super, these enormously powerful bombs—and the behemoth 5000-Hiroshima weapon exploded later by the Soviets in an attempt to intimidate John Kennedy—have not been very attractive to the military establishments of either the United States or the U.S.S.R. The weapons currently in Russian and American arsenals are around 30 Hiroshimas, not thousands. Although they are true hydrogen bombs, they are no more powerful than a large atomic bomb. The military neither needed nor wanted a "doomsday" device. The sole use of such a device would be psychological intimidation of the adversary—but intimidation can be a serious matter in a world with leaders like Joseph Stalin.

On 2 July 1953, Lewis Strauss, a member of the Atomic Energy Commission who had fought bitterly with Oppenheimer over the crash program for the super, became the Commission's chairman. As one of his first acts in power, he ordered removal of all classified material from Oppenheimer's Princeton office. Strauss and many others in Washington were deeply suspicious of Oppenheimer's loyalty. How could a man loyal to America oppose the super effort, as he had before Wheeler's team demonstrated that the Teller–Ulam invention would work? William Borden, who had been chief counsel of Congress's Joint Committee on Atomic